A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 25

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 10^3 / 6^3 = 1000 / 216 **1** = 4.6 times (accept 4.63) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 5 points below. **1** C has the highest velocity (most energy, and energy rises with the cube of velocity), strong reliable winds over water, few land-use conflicts, but long expensive cable and offshore installation / maintenance costs **1** A is windy but on a migration route (bird strikes) and peat bog (foundations / tracks damage hydrology; carbon release) – far from grid – scenic objections **1** B is close to the grid with good wind but near a village (noise, visual objection, shadow flicker) - possible with careful siting **1** D is cheap to connect and land can still be farmed between turbines, but low velocity gives little energy **1** Judgement, eg C or B – turbines spaced 3-5 blade diameters apart to avoid wind shadow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	5
01.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Driven by wind from any direction so no motor to turn them – suitable for turbulent urban sites **1** Generator can be at ground level so a lighter tower **1** Quieter **1** Operate at lower wind velocities **1** Blades more visible so fewer bird deaths **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2
01.4	1 mark for the point below. **1 No gearbox:** fewer breakdowns, less friction so start generating at lower wind speeds, quieter, lighter nacelle (DC converted to AC by an inverter) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 206-209; AQA spec 3.3.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Minimum ozone fell steeply from about 225 DU in 1979 to about 95 DU in the mid 1990s **1** Since then it has fluctuated around 100-115 DU with a slight recovery / no further decline **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (225 - 95) / 225 x 100 **1** = 57.8% (accept 58%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.3	**1 mark each** 1 mark for each of the 4 points below. **1** Stratospheric temperatures over Antarctica are the lowest on Earth so stratospheric clouds / ice crystals form **1** Ice crystal surfaces allow reactions that convert the chlorine released from CFCs into chlorine molecules (Cl_2) during the dark winter **1** The polar vortex (rotating winds) prevents mixing with the rest of the atmosphere, keeping temperatures low and chlorine concentrated **1** When sunlight returns in spring (September) it splits the chlorine molecules into free radicals that destroy ozone – levels recover by December as the vortex breaks down **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4

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Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Ground and satellite UV / ozone measurements show depletion but not the chemistry / correlation is not causation **1** Air samples collected from the stratosphere by helium balloons and high-flying research aircraft detected chlorine and chlorine monoxide, confirming the Rowland-Molina hypothesis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine • **correlation** → a relationship in which the two variables change together • **causation** → one variable actually causing the change in the other **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 110-111; AQA spec 3.2.1.3.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) **1** Most phosphorus compounds have low solubility so little is available in soil water **1** Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil **1** Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mycorrhizal** → a fungus living with plant roots that improves nutrient uptake • **symbiotic** → a close relationship in which both species benefit	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 3 points below. **1** Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) **1** Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment **1** Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates) – harvesting removes phosphorus from farmland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
03.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock **1** Breed crops that absorb phosphate more efficiently **1** Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) **1** Recover phosphate from sewage effluent by iron(III) sulfate precipitation – reduce soil erosion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **mycorrhizal** → a fungus living with plant roots that improves nutrient uptake **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Readings drift so they may not be accurate – calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions **1** pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly **1** If a meter cannot be adjusted, record the percentage error and correct the results **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Same height above the ground **1** No obstacles to air flow – impeller type facing into the wind **1** Axis of rotation horizontal (impeller) or vertical (cup) – same time period / averaging **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling **1** Evaporation cools the wet bulb more when humidity is low – the temperature difference is compared with a table to find humidity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Test strips with reagent bands compared with a colour chart (semi-quantitative) **1 Electronic colourimeter:** reagent added and colour intensity measured at a specific wavelength **1** Ion-selective electrode measuring electrode potential (other ions can interfere) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	
05.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (15.0 - 2.2) / 15.0 x 100 **1** = 85.3% (accept 85%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

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Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Added as an anti-knock agent to smooth combustion and reduce engine wear **1** Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food **1** Lead is a neurotoxin that bioaccumulates – chronic exposure damages the nervous system / brain development in children **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust	3
05.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Replacement of lead water pipes / lead solder with copper / tin-based solder – phosphoric acid dosing of water supplies forming insoluble lead phosphate **1** Phasing out of lead-based paint / safer paint removal **1** Lead-free electrical solder **1** Restrictions on lead fishing weights / shotgun pellets **1** Industrial emission controls **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe **1** This prevents lead dissolving into the water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.5	1 mark for the point below. **1** Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **Critical Group** → the people most at risk / most heavily exposed	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate **1** No need to sample sites that were never likely to be polluted – sampling also checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify **1** So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural pressure of gas above the oil (or dissolved in it) or water beneath forces oil up the production well – pump-jacks ('nodding donkeys') increase flow **1** Pressure falls as oil is removed so the remaining oil no longer flows to the well **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Water or natural gas is pumped down an injection well to maintain the pressure and flow of oil (recovery to about 40%) **1** Captured CO2 can be pumped down instead, increasing oil recovery while storing the CO2 underground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carbon capture** → capturing carbon dioxide from flue gases and storing it underground	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Steam injection (from combustion or solar parabolic concentrators) to heat the oil and reduce its viscosity **1** Controlled underground combustion to heat the oil **1** Detergents / solvents to reduce surface tension so oil flows more easily **1** Bacteria that partially digest heavy oil into lighter oils and produce CO2 to maintain pressure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.4	**1 mark each** 1 mark for each of the 3 points below. **1** Many wells can be drilled from a single platform, reducing land take / number of rigs **1** Deposits under locations where rigs cannot be placed (urban areas, sensitive habitats) can be reached **1** Drilling can follow softer strata (quicker) and target multiple small reservoirs up to 10 km away, increasing total recovery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187-188; AQA spec 3.3.4.1.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - rivers are accessible over a large area with few natural contaminants	1
08.2	**1 mark each** 1 mark for each of the 2 points below. **1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health) **1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange – spray irrigation: low turbidity to avoid blocked pipes and low heavy metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) **1 Disadvantage:** high cost / energy for pumping – ecological impacts on the donor river – transfer of pollutants or invasive species – political conflict **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored **1** Hydrogen has a high energy density so it can be transported / stored for long periods **1** Released by combustion for heat, industrial processes or steam turbines **OR** by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water **1** Can also be used as a vehicle fuel or converted to methane / ammonia **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
09.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks **1** Liquefied at very low temperature - energy for refrigeration and compression **1** Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy **1** Conversion to ammonia - some energy needed but less than the alternatives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging **1** A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **fuel cell** → a cell that produces electricity from hydrogen and oxygen	2
09.4	1 mark for the point below. **1** Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.	
10.1	**1 mark each** 1 mark for each of the 3 points below. **1** Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) **1** They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone **1** The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3

Do not write outside the box

Question	Marking guidance	Marks
10.2	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Provides carbon, oxygen, hydrogen and nitrogen (as N_2, O_2, CO_2, H_2O) for biological molecules **1** Delays the escape of infrared energy, keeping the surface warm **1** Distributes heat and water vapour by winds / ocean currents **1** Provides atmospheric pressure so liquid water can exist **1** Deflects damaging radiation of the solar wind in the upper atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
10.3	1 mark for the point below. **1** Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **stratified** → sampling each area or group in proportion to its size	1

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Troposphere:** enhanced greenhouse effect - anthropogenic CO_2 (fossil fuels, deforestation), methane (rice, livestock, landfill), N_2O, CFCs, tropospheric ozone – more infrared absorbed and re-emitted – aerosols and smoke increase albedo / scatter light – land-use change alters surface albedo and evapotranspiration • **Stratosphere:** CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium – less UV absorbed in the stratosphere, more reaches the surface – stratospheric cooling • **Consequences:** changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions – increased UV damage to organisms • **Difficulties of prediction:** linked processes where one change alters others – feedback (positive and negative) and tipping points – time lags (ocean heat capacity, long residence times of CFCs and CO_2) – natural variability (solar output, volcanoes, Milankovitch cycles) – incomplete understanding of atmospheric processes – limited historical data so proxies needed – models with different assumptions • **Evaluation:** some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain – precautionary principle • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **enhanced greenhouse effect** → extra warming caused by added greenhouse gases • **anthropogenic** → human-caused / caused by human activity • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	In the troposphere human activity enhances the greenhouse effect: fossil fuels and deforestation add CO_2, rice, livestock and landfill add methane, fertilisers add nitrous oxide, and CFCs and tropospheric ozone are new absorbers. More outgoing infrared is absorbed and re-emitted, warming the surface, while smoke and aerosols scatter sunlight and land-use change alters albedo and evapotranspiration. In the stratosphere, CFCs, halons and HCFCs release chlorine and bromine that catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium so that less UV is absorbed there and more reaches the surface, cooling the stratosphere as well.	
	The consequences run through every system the atmosphere's energy drives: climate, thermohaline circulation, the hydrological cycle, sea level and species distributions, plus UV damage to organisms. Predicting them is hard because the processes are linked - one change alters others - and because feedback can amplify (albedo loss, methane release) or damp (cloud cover, photosynthesis) the effect, sometimes past tipping points. Time lags from ocean heat capacity and the long residence times of CO_2 and CFCs, natural variability from solar output, volcanoes and orbital cycles, sparse historical data replaced by proxies, and models built on different assumptions all add uncertainty.	
	Some things are nevertheless well established: the physics of infrared absorption, the chemistry of ozone loss, and the success of the Montreal Protocol. What is uncertain is the scale and timing of regional impacts, which is exactly why the precautionary principle, not certainty, should guide action.	
	Weaker answer - Level 1 (3 marks)	
	Humans release greenhouse gases like carbon dioxide from burning fossil fuels into the troposphere, which traps more heat and causes global warming. CFCs go into the stratosphere and destroy the ozone layer, so more UV reaches the Earth. It is hard to predict what will happen because the atmosphere is complicated and scientists do not know everything about it. There could be feedback effects that make it worse.	
	What makes the difference	
	• Level 1 gives one cause per layer; Level 3 lists several gases and processes and explains the mechanism (absorption and re-emission; catalytic destruction shifting an equilibrium).	
	• 'Difficulties of prediction' is answered in Level 3 with specific reasons (linked processes, feedback, time lags, natural variability, proxy data, models); Level 1 says 'complicated'.	
	• Level 3 evaluates - separating what is certain from what is not - and links this to the precautionary principle.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Attributes:** accuracy, precision, representative samples, reliability, anomalies, validity – hypothesis and null hypothesis – preliminary studies • **Sampling design:** random vs systematic location and avoiding bias – timing (diurnal, seasonal, long-term, weather) – sample size and number and the running mean – standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • **Statistics:** variability, standard deviation and error bars, significance and p-values, correlation vs causation – choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • **Application to Paper 1 topics:** monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	

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Question	Marking guidance	Marks
11.2	**Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Hydrological cycle processes and reservoirs – residence times – dynamic equilibrium – increasing demand (population, affluence, irrigation, industry) • **Human impacts on transfer rates:** deforestation, agriculture, urbanisation, climate change, abstraction – consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • **Case studies of unsustainable exploitation:** Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • **Sustainable strategies:** metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair – new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis – pollution control (sewage, AMD, buffer strips)	25
	• Links to thermohaline circulation and climate – energy costs – circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	

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